

IBSBMS

Developer Documentation

Internship Submission

1. PROJECT OVERVIEW

Project Name:

IBSBMS – Islami Bank Share & Bond Management System

Purpose:

IBSBMS is a web-based shareholder management system designed to manage shareholder account information and control changes through a Maker–Checker approval workflow.

The system supports:

- Viewing shareholder accounts
- Creating new shareholder records
- Modifying existing shareholder records
- Submitting changes for approval
- Checker approval, rejection, or return for modification
- Tracking request history
- Maintaining audit information

The system uses Spring Boot for the backend, Thymeleaf for the web interface, and Oracle Database for persistent storage.

2. TECHNOLOGY STACK

Backend:

- Java 21
- Spring Boot 4.1.1
- Spring MVC
- Spring Data JPA
- Jakarta Validation
- Spring Security

Frontend:

- Thymeleaf

- HTML
- CSS
- Bootstrap 5.3.3
- JavaScript where required

Database:

- Oracle Database
- Oracle JDBC Driver (ojdbc17)

Development Tools:

- IntelliJ IDEA
- Maven
- Git / GitHub
- GitHub Desktop

3. SYSTEM ARCHITECTURE

The application follows a layered architecture.



Main flow:

```
User -> Controller -> Service -> Repository -> Oracle DB
                        |
```

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Business Rules

|

v

Approval Workflow

4. PROJECT STRUCTURE

The main package structure is organized as follows:

com.example.ibsbms

```

|
+-- config
|   +-- SecurityConfig
|
+-- controller
|   +-- HomeController
|   +-- DashboardController
|   +-- ShareholderController
|   +-- ApprovalController
|
+-- dto
|   +-- BasicInfoDto
|   +-- AddressDto
|   +-- BankInfoDto
|   +-- ShareholderCreateRequest
|
+-- entity
|   +-- Shareholder
|   +-- ShareAddress
|   +-- ShareBankInfo
|   +-- ApprovalRequest
|   +-- ApprovalAction
|   +-- ShareholderChangeRequest
|   +-- BusinessAudit
|
+-- repository
|   +-- ShareholderRepository
|   +-- ShareAddressRepository
|   +-- ShareBankInfoRepository
|   +-- ApprovalRequestRepository
|   +-- DashboardRepository
|
+-- service
|   +-- ShareholderService
|   +-- ShareholderWorkflowService
|   +-- ApprovalWorkflowService
|   +-- DashboardService
|   +-- WorkflowIdService
|
+-- resources
|   +-- templates
|       |   +-- shareholder
|       |   +-- approval
|       |   +-- fragments
|       |
|   +-- static
|       +-- css

```

5. USER ROLES

The application uses two main roles.

MAKER

The Maker is responsible for entering and modifying shareholder information.

Maker permissions include:

- View shareholder records
- Create shareholder records
- Modify shareholder records
- Submit requests for approval

- View own requests
- View requests returned for modification
- Resubmit returned requests

CHECKER

The Checker is responsible for reviewing submitted requests.

Checker permissions include:

- View shareholder records
- View pending approvals
- Review request details
- Approve requests
- Reject requests
- Return requests to the Maker for modification
- View rejected requests
- View returned requests

6. SHAREHOLDER MODULE

The Shareholder module manages the final shareholder master data.

The main master tables are:

T_ACCOUNT_SHARE

Stores the primary shareholder/account information.

Examples:

- FOLIO_BO
- CUST_NAME
- FATHER_NAME
- MOTHER_NAME
- PHONE
- EMAIL
- DOB
- NID_NO
- TIN_NO
- SHARES
- BALANCE
- IS_VALID

T_ADDRESS_SHARE

Stores shareholder address information.

Examples:

- FOLIO_BO
- ADD1
- ADD2
- ADD3

- ADD4
- COUNTRY_NAME

T_BANKINFO_SHARE

Stores optional bank information.

Examples:

- FOLIO_BO
- BANK_NAME
- ACC_NO
- BRANCH_NAME
- ROUTING_NO

7. CREATE SHAREHOLDER WORKFLOW

A Maker does not directly create the final approved shareholder

record.

The workflow is:

Maker enters information

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Server creates/reserves Folio/BO

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Create change request

v

Approval request = PENDING_CHECKER

v

Checker reviews request

v v

APPROVE REJECT

v v

Update master Request ends

tables

v

Approved shareholder

The proposal is stored in T_SHAREHOLDER_CHANGE_REQUEST.

The approval state is stored in T_APPROVAL_REQUEST.

Actions such as Maker submission and Checker decisions are recorded
in T_APPROVAL_ACTION.

8. MODIFY SHAREHOLDER WORKFLOW

For an existing approved shareholder, the Maker loads the approved

snapshot and edits the permitted fields.

The existing master data is not immediately overwritten.

Workflow:

Approved shareholder

v

Maker selects Edit

v

Load approved snapshot

v

Maker changes information

v

Create SHAREHOLDER_UPDATE request

|

v

Checker review

+-----+
| |

v v

Approve Return/Reject

| |

v v

Update master Request remains

tables outside approval

|

v

Updated shareholder

The modification request keeps:

OLD_VALUE

Snapshot of the previously approved data.

NEW_VALUE

The Maker's proposed changes.

9. RETURN FOR MODIFICATION

A Checker can return a request when changes are required.

When a request is returned:

- Status becomes RETURNED_FOR_MODIFICATION
- Current stage becomes MAKER
- Checker information is retained
- Return remarks are recorded
- An approval action is created

The Maker can open the returned request and correct the data.

On resubmission:

- The same REQUEST_ID is retained
- The existing OLD_VALUE is preserved
- NEW_VALUE is replaced with the corrected proposal

- Version is incremented
- Status becomes PENDING_CHECKER
- Current stage becomes CHECKER
- Checker assignment is cleared
- A RESUBMITTED action is recorded

This allows the complete request lifecycle to be tracked.

10. APPROVAL DATA MODEL

The approval workflow uses separate tables for the proposal,

approval state, and action history.

T_SHAREHOLDER_CHANGE_REQUEST

Stores the proposed change.

Important information:

- CHANGE_ID
- BUSINESS_REF
- OPERATION_CODE
- OLD_VALUE
- NEW_VALUE
- VERSION

T_APPROVAL_REQUEST

Stores the current approval state.

Important information:

- REQUEST_ID
- BUSINESS_REF
- SOURCE_TYPE
- OPERATION_CODE
- STATUS
- CURRENT_STAGE
- MAKER_ID
- CHECKER_ID
- CREATED_AT
- UPDATED_AT
- DECIDED_AT

T_APPROVAL_ACTION

Stores the history of workflow actions.

Examples:

- SUBMITTED
- APPROVED
- REJECTED

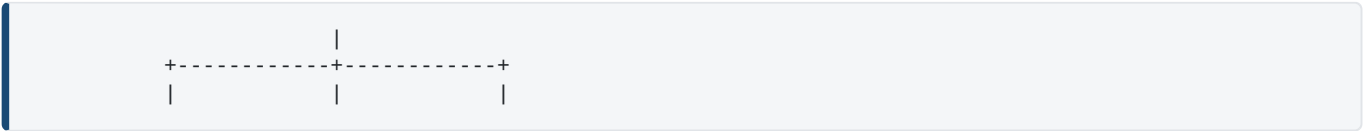
- RETURNED_FOR_MODIFICATION
- RESUBMITTED

T_BUSINESS_AUDIT

Stores business-level audit information for traceability.

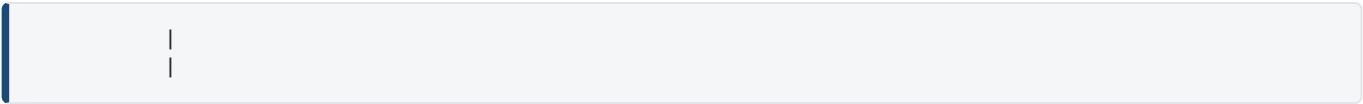
11. DATABASE RELATIONSHIP OVERVIEW

FOLIO_BO



v v v

T_ACCOUNT_SHARE T_ADDRESS_SHARE T_BANKINFO_SHARE



v

T_SHAREHOLDER_CHANGE_REQUEST



v

T_APPROVAL_REQUEST



v

T_APPROVAL_ACTION

Business audit information is maintained separately in:

T_BUSINESS_AUDIT

12. CONTROLLER LAYER

HomeController

Handles the main navigation and shareholder list.

Important routes:

- GET /
- GET /shareholders

DashboardController

Displays dashboard statistics.

Route:

- GET /dashboard

ShareholderController

Handles shareholder creation, modification, viewing, and returned request resubmission.

Important routes:

- GET /shareholders/create
- POST /shareholders/create
- GET /shareholders/{folioBo}/view
- GET /shareholders/{folioBo}/edit
- POST /shareholders/{folioBo}/edit
- GET /shareholders/returned
- GET /shareholders/returned/{requestId}/edit
- POST /shareholders/returned/{requestId}/resubmit

ApprovalController

Handles Checker operations.

Important routes:

- GET /approvals
- GET /approvals/{requestId}
- POST /approvals/{requestId}/approve
- POST /approvals/{requestId}/return
- POST /approvals/{requestId}/reject
- GET /approvals/rejected
- GET /approvals/returned

It also provides Maker request history through:

- GET /my-requests
- GET /my-requests/{requestId}

13. SERVICE LAYER

ShareholderService

Responsible for shareholder-related business operations such as:

- Reading shareholder snapshots
- Building proposal JSON
- Parsing proposal JSON
- Applying an approved shareholder change
- Updating shareholder, address, and bank information

ShareholderWorkflowService

Responsible for submitting and managing shareholder change requests.

Examples:

- Submit create request
- Submit modify request
- Retrieve returned request data
- Resubmit returned request

ApprovalWorkflowService

Responsible for Checker workflow operations.

Examples:

- Retrieve pending requests
- Approve request
- Reject request
- Return request for modification
- Retrieve approval history
- Retrieve decision/return remarks

DashboardService

Retrieves dashboard statistics from the database.

14. REPOSITORY LAYER

Repositories use Spring Data JPA.

Examples:

ShareholderRepository

Provides shareholder CRUD operations and custom queries for the shareholder list and Folio/BO search.

ShareAddressRepository

Provides address data access using Folio/BO.

ShareBankInfoRepository

Provides bank information data access using Folio/BO.

ApprovalRequestRepository

Retrieves:

- Pending Checker requests
- Rejected requests
- Returned requests
- Maker request history

DashboardRepository

Provides database counts for:

- Total shareholders
- Active accounts
- Pending approvals
- Dormant/invalid accounts

15. SECURITY

Spring Security protects application routes based on user roles.

Maker-only operations:

- Create shareholder
- Edit shareholder
- Returned request modification
- Resubmission
- Maker request history

Checker-only operations:

- Approval list
- Approve
- Reject
- Return for modification
- Rejected request list
- Returned request list

Both Maker and Checker can:

- View dashboard
- View shareholder list
- View shareholder details

The current development environment uses in-memory test users.

Production authentication should use the organization's actual authentication/user-management mechanism.

16. DASHBOARD

The dashboard retrieves statistics dynamically from Oracle rather

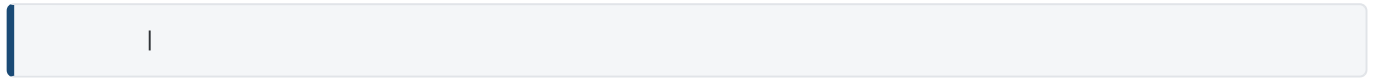
than using hardcoded values.

Displayed statistics include:

- Total Shareholders
- Active Accounts
- Pending Approval
- Dormant Accounts

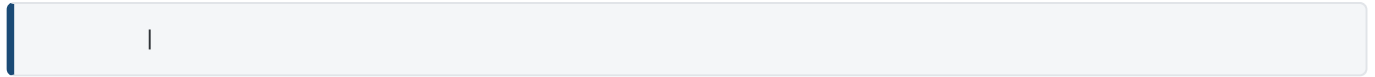
The dashboard flow is:

DashboardController



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DashboardService



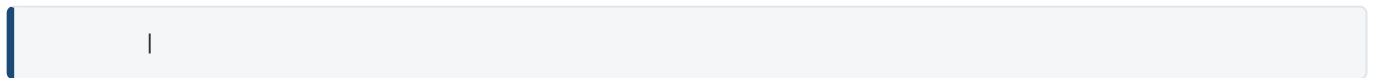
v

DashboardRepository



v

Oracle Database



v

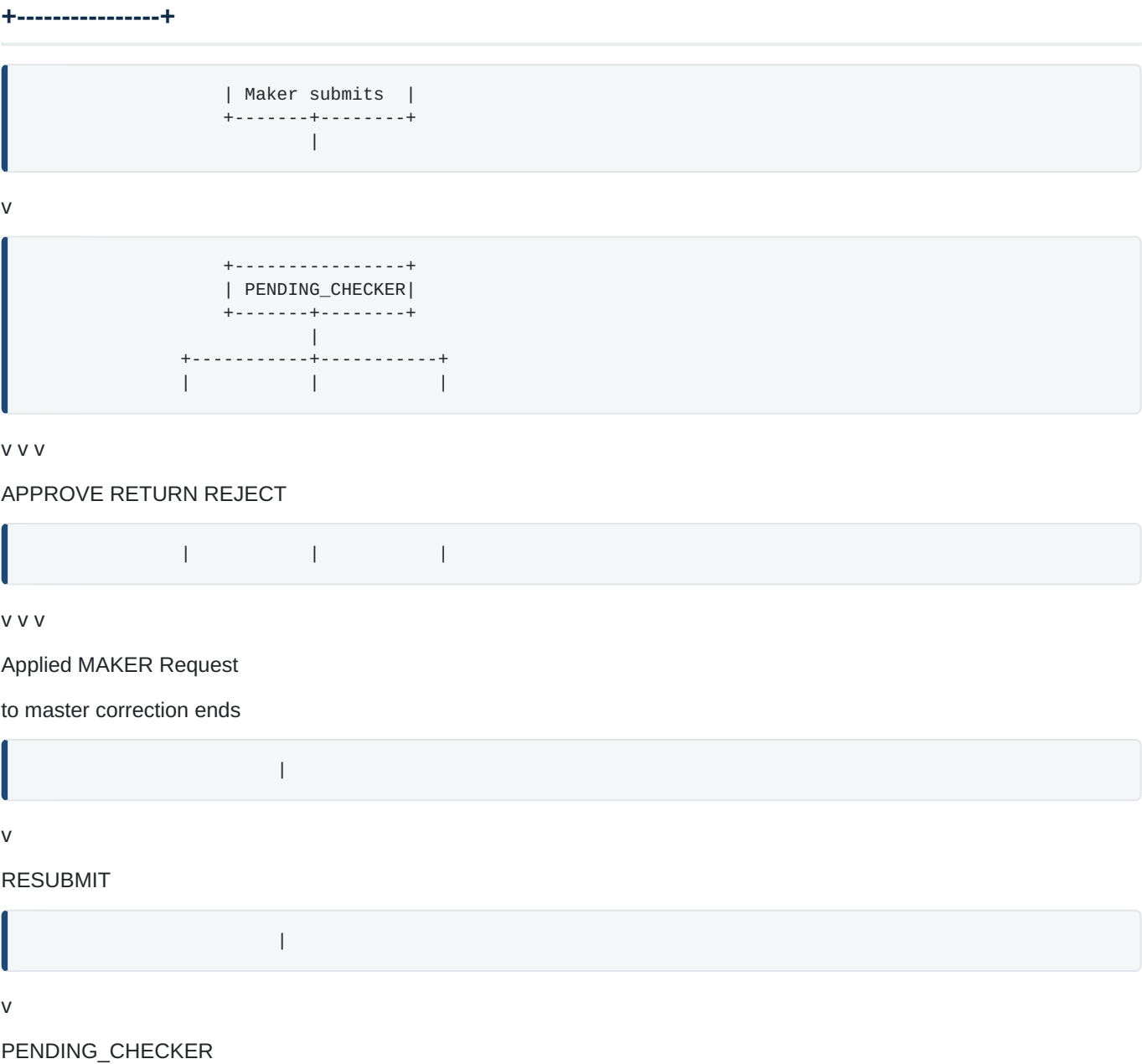
Thymeleaf Dashboard

17. IMPORTANT BUSINESS RULES

1. A Maker submits changes; the Maker does not approve them.

1. A Checker reviews the Maker's request.
1. A Maker cannot act as the Checker for the same request.
1. Final master data should be updated only after approval.
1. Rejected requests are not applied to the master tables.
1. Returned requests can be corrected and resubmitted.
1. The original OLD_VALUE must remain available for modification requests.
1. Shares, Suspense, Bonus, and Balance are not Maker-editable business fields during shareholder modification.
1. Workflow history should remain traceable through approval actions.
1. Business date should be assigned by the server using the Bangladesh business timezone (Asia/Dhaka).

18. TYPICAL REQUEST LIFECYCLE



19. DEVELOPMENT AND DEPLOYMENT NOTES

Prerequisites:

- Java 21
- Maven
- Oracle Database access
- Oracle JDBC driver
- Network/office access required for the Oracle environment

Typical development process:

1. Clone/open the project.

2. Configure the required database connection.
3. Ensure the correct Spring profile is active.
4. Start the Spring Boot application.
5. Access the application through port 8080.
6. Test Maker and Checker workflows separately.
7. Commit changes using Git.
8. Push changes to the team's repository.

The Oracle database may only be reachable when connected to the required office network/IP environment.

20. TESTING CHECKLIST

Before submitting a change, verify:

- ☐ Maker can log in.
- ☐ Checker can log in.
- ☐ Maker can view shareholders.
- ☐ Checker can view shareholders.
- ☐ Maker can create a shareholder request.
- ☐ Created request appears under Checker approvals.
- ☐ Checker can approve the request.
- ☐ Approved data appears in the master tables.
- ☐ Checker can reject a request.
- ☐ Rejected request appears in Rejected Requests.
- ☐ Checker can return a request.
- ☐ Returned request appears for the Maker.
- ☐ Maker can edit and resubmit the returned request.
- ☐ Resubmitted request returns to the Checker queue.
- ☐ Checker Returned Requests disappear after resubmission.
- ☐ Maker cannot access Checker-only endpoints.
- ☐ Checker cannot perform Maker-only edit/create operations.
- ☐ Dashboard values are retrieved from Oracle.
- ☐ Shareholder search and pagination work correctly.

21. CONCLUSION

IBSBMS provides a structured shareholder management solution with

a controlled Maker–Checker workflow. The separation between proposal data, approval state, action history, and final master records provides traceability and helps prevent unauthorized direct changes to shareholder information.

The application is designed using a layered Spring Boot architecture,

with role-based security and Oracle-backed persistence, making it suitable as a foundation for further integration with production banking systems and authentication mechanisms.